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Q.1)Max Marks: 1Subject: Mathematics

The product of 24 and 31 in a number system is 1344 (in the same number system). The number 3111 of this system, when converted to the decimal number system, becomes?

- A) 691
- B) 1086
- C) 381
- D) 406

Correct Option

Solution: (D)

Let the base be n

$$\Rightarrow 24_n \times 31_n = 1344_n$$

$$\Rightarrow (2n+4)(3n+1) = n^3 + 3n^2 + 4n + 4$$

$$\Rightarrow n^3 - 3n^2 - 10n = 0$$

$$\Rightarrow n(n-5)(n+2) = 0$$

$$\Rightarrow n = 5$$

$$(3111)_5 = (406)_{10}$$

Q.2)Max Marks: 1Subject: Mathematics

A number, when divided by 18, leaves a remainder 7.

The same number, when divided by 12, leaves a remainder n . How many values can n take?

Correct Answer

Solution: (2)

The numbers when divided by 18 giving remainder as 7 are

7, 25, 43, 61, 79,

When these numbers are divided by 12 the remainders are 7 and 1

So $n = 2$

Q.3)Max Marks: 1Subject: General Aptitude

How many three digit numbers are divisible by 5 or 9?

- A) 260

- B) 280
- C) 200
- D) 180

Correct Option

Solution: (A)

Three digit numbers divisible by 5 or 9 = Three digit numbers divisible by 5 + Three digit numbers divisible by 9 – Three digit numbers divisible by 5 AND 9 (ie. div by LCM of 5 and 9).

The three digit numbers divisible by 5 = 100, 105, 110....995

The sequence given is in A.P with common difference 5. Let 995 be the n th term of the A.P, then

$$995 = 100 + (n - 1)5 = 100 + 5n - 5$$

Thus, $n = 180$ -----(1)

The three digit numbers divisible by 9 = 108, 118, ... 999

The sequence given is in A.P with common difference 9. Let 999 be the p th term of the A.P, then

$$999 = 108 + (p - 1)9 = 108 + 9p - 9$$

Thus, $p = 100$ ----- (2)

The three digit numbers divisible by 45 = 135, 180, ...990

The sequence given is in A.P with common difference 45. Let 990 be the q th term of the A.P, then

$$990 = 135 + (q - 1)45 = 135 + 45q - 45$$

Thus, $q = 20$ ----- (3)

Thus, from (1), (2) and (3) the three digit numbers divisible by 5 or 9 = $180 + 100 - 20$

= **260**

Q.4)Max Marks: 1Subject: Mathematics

The H.C.F of two numbers, each having three digits, is 17 and their L.C.M. is 714. The sum of the numbers will be

Correct Answer

Solution: (221)

Let two numbers be $17x$ and $17y$ respectively, as 17 is the HCF of these numbers.

Now, LCM of $17x$ and $17y$ is $17xy$

and, according to the question LCM of these two numbers is 714

$$\Rightarrow 17xy = 714$$

$$\Rightarrow xy = 42 = 7 \times 6$$

Now we have two possible scenarios, either,

$$x=7 \text{ and } y=6 \text{ ----(1)}$$

or,

$$x=6 \text{ and } y=7 \text{ ----(2)}$$

In the first scenario, the numbers will be, $17(7)$ and $17(6)$ i.e. 119, 102

In the second scenario, the numbers will be, $17(6)$ and $17(7)$ i.e.

102, 119

The required sum would be 221.

Q.5) Max Marks: 1 Subject: Mathematics

What is the unit digit of $(17^3)^4 - 1973^{3^2}$

- A) 2
- B) 4
- C) 6
- D) 8

Correct Option

Solution: (A)

1. $7^1=7$ (last digit is 7)
2. $7^2=9$ (last digit is 9)
3. $7^3=3$ (last digit is 3)
4. $7^4=1$ (last digit is 1)
5. $7^5=7$ (last digit is 7 again!)
- ...

1. $3^1=3$ (last digit is 3)
2. $3^2=9$ (last digit is 9)
3. $3^3=27$ (last digit is 7)
4. $3^4=81$ (last digit is 1)
5. $3^5=243$ (last digit is 3 again!)
- ...

Thus the units digit of 7^{12} will be 1 (4th in pattern, as 12 is a multiple of cyclicity number 4) and the units digit of 3^9 will be 3 (first in pattern, as $9=4*2+1$).

So, we have that the units digit of $(17^3)^4 = 17^{12}$ is 1 and the units digit of $1973^3 \cdot 3^2 = 1973^9$ is 3. We will get negative value with 2 in the units place as 1973^9 is greater than 7^{12} .

Q.6)Max Marks: 1Subject: Mathematics

A two digit number A in base 11 is one-third of the number formed by reversing its digits when considered in base 19. How many such numbers are possible?

- A) 5
- B) 6
- C) 8
- D) 9

Correct Option

Solution: (A)

Let the number be xy

$$\Rightarrow (xy)_{11} = \frac{1}{3}(yx)_{19}$$

$$\Rightarrow 11x + y = \frac{1}{3}(19y + x)$$

$$\Rightarrow 33x + 3y = 19y + x$$

$$\Rightarrow 32x = 16y$$

$$\Rightarrow 2x = y$$

So, (x,y) can be $(2,1), (4,2), (6,3), (8,4), (10,5)$

Numbers such as 10,11,12...etc are represented as alphabets A, B, C etc. in higher bases. Thus, $(10,5)$ also satisfies our requirements.

Note that 12 is not a single digit number in base 11 and so $(12,6)$ is not a valid pair.

Therefore, the answer is 5.

Q.7)Max Marks: 1Subject: Mathematics

In class A, the ratio of boys to girls is 2 : 3. In class B the ratio of boys to girls is 4 : 5. If the ratio of boys to girls in both classes put together is 3 : 4, If the ratio of the number of girls in class A to number of girls in class B is a:b where a and b are co-primes, find the value of a+b ?

Correct Answer

Solution: (8)

Let us assume class A has 2x boys and 3x girls, class B has 4y boys and 5y girls.

$$\frac{2x+4y}{3x+5y} = \frac{3}{4}$$

$$8x + 16y = 9x + 15y$$

$$x = y$$

$$\text{Required ratio} = \frac{3x}{5y} \text{ (since } x = y \text{)} = \frac{3}{5}$$

Q.8)Max Marks: 1Subject: Mathematics

Sum of two numbers x, y is 1050. What is the maximum value of the HCF(x,y)?

- A) 350
- B) 700
- C) 1050
- D) 525

Correct Option

Solution: (D)

If x = 350 and y = 700 then HCF is 350

If x = 525 and y = 525 then HCF is 525 which is maximum

Q.9)Max Marks: 1Subject: Mathematics

Find the remainder when $1! + 2! + 3! + 4! + 5! + \dots + 100!$ is divided by 24 ?

Correct Answer

Solution: (9)

$$(1 + 2 + 6 + 0 + 0 + \dots) / 24 = 9$$

From 4! onwards all the terms leave a remainder 0 when divided by 24.

Q.10)Max Marks: 1Subject: Mathematics

The next term of the *arithmetic progression* $\sqrt{18}$, $\sqrt{50}$, $\sqrt{98}$ is ?

A) $\sqrt{146}$

B) $\sqrt{128}$

C) $\sqrt{162}$

D) $\sqrt{200}$

Correct Option

Solution: (C)

$$\sqrt{18} = \sqrt{9 \times 2} = 3\sqrt{2}$$

$$\sqrt{50} = \sqrt{25 \times 2} = 5\sqrt{2}$$

$$\sqrt{98} = \sqrt{49 \times 2} = 7\sqrt{2}$$

$$\text{Common difference} = 2\sqrt{2}$$

$$\text{Next term will be } 7\sqrt{2} + 2\sqrt{2} = 9\sqrt{2} = \sqrt{81 \times 2} = \sqrt{162}$$

Q.11)Max Marks: 2Subject: Mathematics

Rs 4830 is divided among Abhishek, Dishant and Prashant such that if Abhishek's share diminishes by Rs 5, Dishant's share diminishes by Rs 10 and Prashant's share diminishes by Rs 15, their shares will be in the ratio 5:4:3. Find the Dishant's original share.

A) 1600

B) 1590

C) 1610

D) 2019

Correct Option

Solution: (C)

Let actual share of Abhishek, Dishant and Prashant be A, D , P respectively.

$$A+D+P=4830$$

Hence, A's, D's and P's share are diminished by Rs 5, Rs 10 and Rs 15, their net share will be Rs.4800.

$$\text{Dishant's diminished share} = (4/12) \times 4800 = \text{Rs } 1600$$

$$\text{Hence, Dishant actual share} = \text{Rs } 1600 + \text{Rs } 10 = \text{Rs } 1610.$$

Q.12)Max Marks: 2Subject: Mathematics

The traffic lights at three different road crossings change after every 48 sec; 72 sec; and 108 sec respectively. If they all change simultaneously at 8:20:00 hrs, then they will again change simultaneously at

- A) 8:27:12 Hrs
- B) 8:27:24 Hrs
- C) 8:27:36 Hrs
- D) 8:27:48 Hrs

Correct Option

Solution: (A)

The change of interval =(LCM of 48,72,108)sec.=432.

So, for every 432 seconds i.e.7 min. 12 sec. the lights will change.

Add 7 min.12 sec.to 8:20:00 Hrs.i.e.8:27:12 Hrs.

Q.13)Max Marks: 2Subject: Mathematics

6 different sweet varieties of count 32, 216, 136, 88, 184, 120 were ordered for a particular occasion. They need to be packed in such a way that each box has the same variety of sweet and the number of sweets in each box is also the same. What is the minimum number of boxes required to pack?

- A) 12

- B) 96
- C) 48
- D) 97

Correct Option

Solution: (D)

All sweets need to be packed and each box has the same variety.

This implies the number of sweets in each box should be HCF of different count of sweets HCF of 32, 216, 136, 88, 184, 120 = 8

Minimum number of boxes = $(32 + 216 + 136 + 88 + 184 + 120) / 8 = 97$

Q.14)Max Marks: 2Subject: Mathematics

. For a positive integer n , let P_n denote the product of the digits of n and S_n denote the sum of the digits of n . The number of integers between 10 and 100 for which $P_n + S_n = n$?

- A) 2
- B) 9
- C) 10
- D) 12

Correct Option

Solution: (B)

Let the two digit number be 'ab'

Given, $a \times b + a + b = 10a + b$

$a \times b = 9a$

$b = 9$

Therefore, for $b = 9$, the above condition satisfies.

So all the two digit numbers for which units digit is 9 are our solutions.

They range from 19 to 99.

Total 9 numbers.

Q.15)Max Marks: 2Subject: Mathematics

A fruit seller has oranges, apples and guavas in the ratio 2:5:8. The number of apples is more than the number of oranges by a number that is a multiple of both 6 and 8. What is the minimum number of fruits in his shop?

- A) 240
- B) 360
- C) 120
- D) 90

Correct Option

Solution: (C)

Let the number of fruits be $2k$, $5k$ and $8k$.

Given, $5k - 2k = \text{multiple of 6 and 8}$

LCM of 6 and 8 is 24

Let's say $5k - 2k = 24n$

$$\Rightarrow 3k = 24n$$

For k to be a natural number and have minimum value, n should be equal to 1

$$\Rightarrow 3k = 24 \text{ Or } k = 8$$

Hence, the number of fruits = $2k + 5k + 8k = 15 * 8 = 120$